

Phase2ML

April 26, 2026

1 Initialization

```
[41]: # Initialization
      #import pyspark as spark
      from pyspark.sql import SparkSession, functions

      # incase the spark context already exists

      spark = SparkSession \
          .builder \
          .appName("Project2") \
          .getOrCreate()
```

```
[42]: # Global Constants
      random_seed = 42

      path_to_file_local = "games_cleaned.csv"
      path_to_file_hadoop = "hdfs://localhost:9000/steam-data/games_cleaned.csv"
```

```
[43]: from pyspark.sql.types import *

      # TODO: Finish this... I'm too tired to finish this tonight
      types = StructType([
          StructField('AppId', IntegerType(), True),
          StructField('Name', StringType(), True),
          StructField('Release date', StringType(), True),
          StructField('Estimated owners', StringType(), True),
          StructField('Peak CCU', IntegerType(), True),
          StructField('Required age', IntegerType(), True),
          StructField('Price', DoubleType(), True),
          StructField('DiscountDLC count', IntegerType(), True),
          StructField('Supported languages', StringType(), True),
          StructField('Full audio languages', StringType(), True),
          StructField('Windows', IntegerType(), True),
          StructField('Mac', IntegerType(), True),
          StructField('Linux', IntegerType(), True),
```

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StructField('Metacritic score', DoubleType(), True),
StructField('User score', DoubleType(), True),
StructField('Positive', IntegerType(), True),
StructField('Negative', IntegerType(), True),
StructField('Average playtime forever', DoubleType(), True),
StructField('Average playtime two weeks', DoubleType(), True),
StructField('Median playtime forever', DoubleType(), True),
StructField('Median playtime two weeks', DoubleType(), True),
StructField('Developers', StringType(), True),
StructField('Publishers', StringType(), True),
StructField('Categories', StringType(), True),
StructField('Genres', StringType(), True),
StructField('Tags', StringType(), True),
StructField('estimated_owners_mid', IntegerType(), True),
StructField('estimated_revenue', DoubleType(), True),
StructField('num_languages', DoubleType(), True)
])

data = spark.read.schema(types).csv(path_to_file_local, header=True)
data.show()

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+
| AppId|                Name|Release date|Estimated owners|Peak
CCU|Required age|Price|DiscountDLC count| Supported languages|Full audio
languages|Windows| Mac|Linux|Metacritic score|User
score|Positive|Negative|Average playtime forever|Average playtime two
weeks|Median playtime forever|Median playtime two weeks|      Developers|
Publishers|          Categories|          Genres|
Tags|estimated_owners_mid|estimated_revenue|num_languages|
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+
|2539430|      Black Dragon Mage...| 2023-08-01|      0 - 0|      0|
0| 0.0|      0|      []|      []|
NULL|NULL| NULL|      0.0| 0.0| 0| 0|
0.0|      0.0|      0.0|
0.0|      None|      None|      ['None']|

```

['None']	['None']	NULL	0.0
0.0			
496350	Supipara - Chapte...	2016-07-29	0 - 20000 0
0 5.24	65	['English']	[]
NULL NULL NULL	0.0	0.0 252	3
8.0	0.0	8.0	
0.0	minori	MangaGamer	['Single-player',...
['Adventure']	['Adventure', 'Vi...	NULL	52400.0
1.0			
1034400	Mystery Solitaire...	2019-05-06	0 - 20000 0
0 4.99	0	['English', 'Fren...	[]
NULL NULL NULL	0.0	0.0 21	3
0.0	0.0	0.0	
0.0	Somer Games	8floor	['Single-player',...
['Casual']	['Casual', 'Card ...	NULL	49900.0
4.0			
3292190	- Vtube...	2024-10-31	0 - 20000 1
0 8.99	0	['Korean']	['Korean']
NULL NULL NULL	0.0	0.0 0	0
0.0	0.0	0.0	
0.0		['Single-player',...	['Casual', 'Indie...
['None']	NULL	89900.0	1.0
3631080	Maze Quest VR	2025-04-24	0 - 20000 0
0 4.99	0	['English']	['English']
NULL NULL NULL	0.0	0.0 0	0
0.0	0.0	0.0	
0.0 Reality Expanded LLC	Reality Expanded LLC	['Single-player',...	['Action', 'Early...
1.0	['None']	NULL	49900.0
1654170	Agony VR	2023-04-05	0 - 20000 0
0 13.99	0	['English', 'Fren...	['English', 'Turk...
NULL NULL NULL	0.0	0.0 0	0
0.0	0.0	0.0	
0.0	Ignibit	Ignibit,Madmind S...	['Single-player',...
'Adven...	['None']	NULL	139900.0
9.0			
1934300	Armored Brigade II	2025-04-08	0 - 20000 8
0 35.99	10	['English']	[]
NULL NULL NULL	0.0	0.0 117	13
675.0	0.0	1253.0	
0.0	Veitikka Studios	Matrix Games	['Single-player',...
359900.0	1.0	['Simulation', 'S...	['Simulation', 'S...
700340	Galacatraz: Eject...	2018-02-01	0 - 20000 0
0 0.99	0	['English']	['English']
NULL NULL NULL	0.0	0.0 0	0
0.0	0.0	0.0	
0.0	Summer Bear Studio	Summer Bear Studio	['Single-player',...
			['Action',

'Indie']	['None']	NULL	9900.0
1.0			
1157670	Hepta Beats	2021-05-07	0 - 20000 0
0 0.99	0 ['English', 'Fren...	['English', 'Fren...	
NULL NULL NULL	0.0	0.0 28	4
0.0	0.0	0.0	
0.0	Matheus Bigatão	Matheus Bigatão	['Single-player',... ['Casual',
'Indie']	['Precision Platf...	NULL	9900.0
29.0			
1540330	MUMBA IV: Egypt J...	2021-12-13	0 - 20000 1
0 0.59	0 ['English', 'Fren...		[]
NULL NULL NULL	0.0	0.0 19	6
0.0	0.0	0.0	
0.0	Airem	Airem	['Single-player',... ['Action',
'Casua...	['Casual', 'Actio...	NULL	5900.0
30.0			
1946890	Codename: Warlock...	2022-03-25	0 - 0 0
0 0.0	0	[]	[]
NULL NULL NULL	0.0	0.0 0	0
0.0	0.0	0.0	
0.0	None	None	['None']
['None']	['None']	NULL	0.0
0.0			
1108640	OMNIMUS	2019-09-25	0 - 20000 0
0 0.99	0 ['English', 'Russ...		[]
NULL NULL NULL	0.0	0.0 47	6
0.0	0.0	0.0	
0.0	Angry games	Total	['Single-player',... ['Adventure',
'In...	['Adventure', 'In...	NULL	9900.0
2.0			
1025440	Fantasy General II	2019-09-05	100000 - 200000 13
0 13.99	0 ['English', 'Germ...		[]
NULL NULL NULL	80.0	0.0 1025	210
489.0	0.0	52.0	
0.0	Owned by Gravity	Slitherine Ltd.	['Single-player',...
['Strategy']	['Strategy', 'Tur...	NULL	2098500.0
6.0			
2113450	Forests, Fields a...	2023-04-05	0 - 20000 0
0 1.49	70 ['English', 'Russ...		[]
NULL NULL NULL	0.0	0.0 64	2
0.0	0.0	0.0	
0.0	9 Eyes Game Studio	9 Eyes Game Studi...	['Single-player',... ['Casual',
'Indie']	['Casual', 'Strat...	NULL	14900.0
2.0			
2922740	DogDogDog	2024-04-19	0 - 20000 0
0 1.35	66 ['English', 'Simp...		[]
NULL NULL NULL	0.0	0.0 34	10
0.0	0.0	0.0	

0.0	LuuuLuuuL	LuuuLuuuL	['Single-player',...	['Adventure',
'Ca...	['Dog', 'Cute', '...	NULL	13500.0	
4.0				
1282900	Bacon Quest - Lov...	2020-04-29	0 - 20000	0
0 0.99	0	['English']	[]	
NULL NULL	NULL	0.0	0.0	0
0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	
0.0	Witching Hour Ent...	Witching Hour Ent...	['Single-player',...	
['RPG']	['None']	NULL	9900.0	
1.0				
2517090	Challenge from th...	2023-08-10	0 - 20000	0
0 1.4	0	['English', 'Japa...	['Japanese', 'Kor...	
NULL NULL	NULL	0.0	0.0	2
0.0	0.0	0.0	0.0	
0.0	ZENO	ZENO	['Single-player',...	
['Action']	['Action', 'Shoot...	NULL	14000.0	
13.0				
328460	Redline	2014-10-23	0 - 20000	0
0 1.04	0	['English']	[]	
NULL NULL	NULL	0.0	0.0	74
203.0	0.0	0.0	203.0	35
0.0	Accolade, Inc.	Atari,Nightdive S...	['Single-player',...	['Action',
'Racing']	['Action', 'Racin...	NULL	10400.0	
1.0				
2260230	Midnight Dash	2023-01-09	0 - 0	0
0 0.0	0	['English']	['English']	
NULL NULL	NULL	0.0	0.0	0
0.0	0.0	0.0	0.0	
0.0	Saku Hållfast	Saku Hållfast	['Single-player',...	['Action',
'Adven...	['None']	NULL	0.0	
1.0				
2403000	Fantasy Amusement...	2023-07-28	0 - 20000	1
0 5.99	0	['English', 'Japa...	['Japanese', 'Ara...	
NULL NULL	NULL	0.0	0.0	0
88.0	0.0	0.0	96.0	0
0.0	Cat Paw Game	Shark Shark Game	['Single-player',...	
['Casual']	['None']	NULL	59900.0	
18.0				

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2 Data Preparation

2.0.1 Train Test Split

```
[44]: from pyspark.ml.feature import VectorAssembler

# Note: We probably need to translate some of the strings to integer
# enumerations...
# but until then....

string_columns = [c for c, t in data.dtypes if t == 'string']
#string_columns = [c.name for c in types.fields if c.typeName == 'str']
svm_data = data.drop(*string_columns)

assembler = VectorAssembler(
    inputCols=svm_data.columns,
    outputCol='features',
    handleInvalid='keep'
)

data_separted = assembler.transform(svm_data.fillna(0) \
    .withColumn('sold_well', functions.when(functions.col('estimated_revenue') >=
    500000, 1).otherwise(0))
    ).drop('estimated_revenue')

data_separted.show()
```

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| AppId|Peak CCU|Required age|Price|DiscountDLC
count|Windows|Mac|Linux|Metacritic score|User score|Positive|Negative|Average
playtime forever|Average playtime two weeks|Median playtime forever|Median
playtime two weeks|estimated_owners_mid|num_languages|sold_well|
features|
+-----+-----+-----+-----+-----+-----+-----+-----+
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|2539430|      0|      0| 0.0|      0|      0| 0| 0|      0|
0.0|      0.0|      0|      0|      0.0|
0.0|      0.0|
0.0|      0|(19,[0],[2539430.0])|
| 496350|      0|      0| 5.24|      65|      0| 0|      0|
0.0|      0.0|      252|      3|      8.0|
0.0|      8.0|      0.0|      0|
1.0|      0|(19,[0,3,4,10,11,...|
|1034400|      0|      0| 4.99|      0|      0| 0|      0|
```

0.0	0.0	21	3
0.0		0.0	
4.0		0 (19,[0,3,10,11,17...	
3292190		1	0 8.99
0.0	0.0	0	0
0.0		0.0	
1.0		0 (19,[0,1,3,17,18]...	
3631080		0	0 4.99
0.0	0.0	0	0
0.0		0.0	
1.0		0 (19,[0,3,17,18],[...	
1654170		0	0 13.99
0.0	0.0	0	0
0.0		0.0	
9.0		0 (19,[0,3,17,18],[...	
1934300		8	0 35.99
0.0	0.0	117	13
0.0		1253.0	
1.0		0 (19,[0,1,3,4,10,1...	
700340		0	0 0.99
0.0	0.0	0	0
0.0		0.0	
1.0		0 (19,[0,3,17,18],[...	
1157670		0	0 0.99
0.0	0.0	28	4
0.0		0.0	
29.0		0 (19,[0,3,10,11,17...	
1540330		1	0 0.59
0.0	0.0	19	6
0.0		0.0	
30.0		0 (19,[0,1,3,10,11,...	
1946890		0	0 0.0
0.0	0.0	0	0
0.0		0.0	
0.0		0 (19,[0],[1946890.0])	
1108640		0	0 0.99
0.0	0.0	47	6
0.0		0.0	
2.0		0 (19,[0,3,10,11,17...	
1025440		13	0 13.99
80.0	0.0	1025	210
0.0		52.0	
6.0		1 (19,[0,1,3,8,10,1...	
2113450		0	0 1.49
0.0	0.0	64	2
0.0		0.0	
2.0		0 (19,[0,3,4,10,11,...	
2922740		0	0 1.35

0.0			
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	10	0	0
	675.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	0.0		
0.0			0
	0	0	0
	489.0		
0.0			0
	70	0	0
	0.0		
0.0			0
	66	0	0

```

0.0|      0.0|      34|      10|      0.0|
0.0|      0.0|
4.0|      0|(19,[0,3,4,10,11,...|
|1282900|      0|      0| 0.99|
0.0|      0.0|      0|      0|
0.0|      0.0|
1.0|      0|(19,[0,3,17,18],[...|
|2517090|      0|      0| 1.4|
0.0|      0.0|      2|      0|
0.0|      0.0|
13.0|      0|(19,[0,3,10,17,18...|
| 328460|      0|      0| 1.04|
0.0|      0.0|      74|      35|
0.0|      203.0|
1.0|      0|(19,[0,3,10,11,12...|
|2260230|      0|      0| 0.0|
0.0|      0.0|      0|      0|
0.0|      0.0|
1.0|      0|(19,[0,18],[22602...|
|2403000|      1|      0| 5.99|
0.0|      0.0|      0|      0|
0.0|      96.0|
18.0|      0|(19,[0,1,3,12,14,...|
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only showing top 20 rows

```
[45]: train, test = data_separeted.randomSplit([0.7, 0.3], seed=random_seed)
train.show()
```

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|AppId|Peak CCU|Required age|Price|DiscountDLC
count|Windows|Mac|Linux|Metacritic score|User score|Positive|Negative|Average
playtime forever|Average playtime two weeks|Median playtime forever|Median
playtime two weeks|estimated_owners_mid|num_languages|sold_well|
features|
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| 420|      20|      0| 0.0|      0|      0| 0| 0|
90.0|      0.0| 40556| 1226|      0.0|
0.0|      0.0|      0.0|      0.0|      0|

```


17.0	0 (19,[0,1,8,10,11,...
570	623941 0 0.0
90.0	0.0 2037143 461826
1427.0	1155.0
28.0	0 (19,[0,1,8,10,11,...
1530	1 0 2.39
76.0	0.0 478 107
0.0	30.0
5.0	0 (19,[0,1,3,8,10,1...
2210	18 17 4.94
81.0	0.0 3001 423
46.0	66.0
1.0	1 [2210.0,18.0,17.0...
2310	82 17 3.99
94.0	0.0 13926 474
76.0	97.0
0.0	0 (19,[0,1,2,3,8,10...
2710	12 0 1.19
82.0	0.0 511 56
0.0	172.0
1.0	0 (19,[0,1,3,4,8,10...
3520	1 0 0.99
0.0	0.0 55 11
0.0	609.0
1.0	0 (19,[0,1,3,4,10,1...
3610	5 0 0.99
0.0	0.0 43 11
0.0	13.0
1.0	0 (19,[0,1,3,10,11,...
6030	40 0 4.99
89.0	0.0 4281 413
0.0	117.0
5.0	1 (19,[0,1,3,4,8,10...
7860	0 17 0.74
63.0	0.0 1181 554
0.0	119.0
9.0	0 (19,[0,2,3,4,8,10...
8800	1395 0 5.99
86.0	0.0 4423 169
1661.0	258.0
5.0	1 [8800.0,1395.0,0...
8880	1 0 1.24
90.0	0.0 292 230
0.0	78.0
1.0	0 (19,[0,1,3,8,10,1...
10240	0 0 2.79
0.0	0.0 21 12
0.0	82.0

0	0	0	0
54064.0			
860.0			0
0	0	0	0
74.0			
0.0			0
0	0	0	0
160.0			
46.0			0
0	0	0	0
323.0			
76.0			0
60	0	0	0
594.0			
0.0			0
80	0	0	0
609.0			
0.0			0
0	0	0	0
13.0			
0.0			0
50	0	0	0
392.0			
0.0			0
85	0	0	0
268.0			
0.0			0
80	0	0	0
2378.0			
1661.0			0
0	0	0	0
629.0			
0.0			0
0	0	0	0
82.0			
0.0			0

1.0	0 (19,[0,3,10,11,12...				
12380	0	0	3.99	80	0 0 0
0.0	0.0	0	0	1.0	
0.0		1.0		0.0	0
1.0	0 (19,[0,3,4,12,14,...				
12640	6	0	1.99	0	0 0 0
75.0	0.0	592	143	1018.0	
9.0		254.0		9.0	0
3.0	0 [12640.0,6.0,0.0,...				
12690	0	0	4.49	0	0 0 0
0.0	0.0	277	55	15.0	
0.0		15.0		0.0	0
1.0	0 (19,[0,3,10,11,12...				
13570	26	17	2.49	0	0 0 0
92.0	0.0	4785	348	523.0	
0.0		266.0		0.0	0
5.0	1 (19,[0,1,2,3,8,10...				
15800	0	0	9.99	0	0 0 0
0.0	0.0	1	2	0.0	
0.0		0.0		0.0	0
1.0	0 (19,[0,3,10,11,17...				
16200	2	0	9.99	0	0 0 0
0.0	0.0	58	9	807.0	
0.0		1026.0		0.0	0
1.0	0 (19,[0,1,3,10,11,...				
18460	0	0	9.99	0	0 0 0
78.0	0.0	1	1	48.0	
0.0		48.0		0.0	0
1.0	1 (19,[0,3,8,10,11,...				

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only showing top 20 rows

```
[46]: # SVM for determining if a game will sell well
from pyspark.ml.classification import LinearSVC
svm = LinearSVC(labelCol='sold_well', featuresCol='features')

# default training
model =svm.fit(train)
```

```
[47]: prediction = model.transform(test)
prediction.show()
```

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|AppId|Peak CCU|Required age|Price|DiscountDLC
count|Windows|Mac|Linux|Metacritic score|User score|Positive|Negative|Average
playtime forever|Average playtime two weeks|Median playtime forever|Median
playtime two weeks|estimated_owners_mid|num_languages|sold_well|
features|      rawPrediction|prediction|
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| 1250|      224|      17| 4.99|      75|      0| 0| 0|
72.0|      0.0| 64024| 3448|      1789.0|
2117.0|      284.0|      4011.0|      0|
8.0|      1|[1250.0,224.0,17...|[-1152.4763169724...|      1.0|
| 2545|      0|      0| 4.99|      0|      0| 0| 0|
0.0|      0.0| 204| 136|      33.0|
0.0|      41.0|      0.0|      0|
748500.0|      0|(19,[0,3,10,11,12...|[25.8288602838525...|      0.0|
| 3300|      41|      0| 0.99|      0|      0| 0| 0|
0.0|      0.0| 601| 35|      637.0|
0.0|      1075.0|      0.0|      0|
5.0|      0|(19,[0,1,3,10,11,...|[28.9947784572119...|      0.0|
| 3510|      4|      0| 0.99|      0|      0| 0| 0|
0.0|      0.0| 70| 17|      0.0|
0.0|      0.0|      0.0|      0.0|      0|
1.0|      0|(19,[0,1,3,10,11,...|[31.6637951181925...|      0.0|
| 6840|      6|      0| 2.49|      50|      0| 0| 0|
72.0|      0.0| 681| 407|      100.0|
0.0|      18.0|      0.0|      0|
1.0|      1|(19,[0,1,3,4,8,10...|[-92.805286630781...|      1.0|
| 7200|      75|      0|14.99|      0|      0| 0| 0|
80.0|      0.0| 2482| 136|      5688.0|
0.0|      463.0|      0.0|      0|
15.0|      1|(19,[0,1,3,8,10,1...|[-322.11930651011...|      1.0|
| 7620|      86|      0| 1.24|      0|      0| 0| 0|
89.0|      0.0| 1162| 133|      1392.0|
0.0|      1652.0|      0.0|      0|
1.0|      0|(19,[0,1,3,8,10,1...|[4.56366004298386...|      0.0|
| 9480|      95|      17| 1.49|      0|      0| 0| 0|
72.0|      0.0| 15420| 5046|      688.0|
530.0|      123.0|      530.0|      0|
9.0|      1|[9480.0,95.0,17.0...|[-320.06177093443...|      1.0|
|11040|      5|      0| 0.99|      90|      0| 0| 0|
71.0|      0.0| 468| 185|      286.0|
0.0|      241.0|      0.0|      0|
5.0|      0|(19,[0,1,3,4,8,10...|[23.9424024374235...|      0.0|
|12520|      3|      0| 4.49|      0|      0| 0| 0|

```

0.0	0.0	360	134	332.0	
0.0		295.0		0.0	0
1.0	0 (19,[0,1,3,10,11,...	[11.2072381113442...		0.0	
12590	0	0 4.49		0 0 0	0
0.0	0.0	35	29	15.0	
0.0		15.0		0.0	0
1.0	0 (19,[0,3,10,11,12...	[23.4101214402596...		0.0	
13600	26	0 1.99		0 0 0	0
89.0	0.0	3844	637	252.0	
146.0		88.0		146.0	0
5.0	1 [13600.0,26.0,0.0...	[-67.384077122014...		1.0	
15000	2	17 2.49		75 0 0	0
59.0	0.0	293	190	21.0	
0.0		32.0		0.0	0
1.0	0 [15000.0,2.0,17.0...	[8.86474289062303...		0.0	
15160	1	0 1.99		0 0 0	0
0.0	0.0	157	52	0.0	
0.0		0.0		0.0	0
3.0	0 (19,[0,1,3,10,11,...	[32.6823750527913...		0.0	
15920	0	0 3.29		0 0 0	0
0.0	0.0	18	28	146.0	
0.0		168.0		0.0	0
3.0	0 (19,[0,3,10,11,12...	[31.8240749228465...		0.0	
16300	0	0 9.99		0 0 0	0
77.0	0.0	110	6	15.0	
0.0		28.0		0.0	0
35000.0	0 (19,[0,3,8,10,11,...	[34.2590660321419...		0.0	
18000	0	17 7.99		0 0 0	0
0.0	0.0	451	58	376.0	
0.0		139.0		0.0	0
1.0	1 (19,[0,2,3,10,11,...	[-47.087132417505...		1.0	
23455	0	0 2.44		0 0 0	0
0.0	0.0	0	0	0.0	
0.0		0.0		0.0	0
3.0	0 (19,[0,3,17,18],[...	[32.3858889390069...		0.0	
32310	5	0 2.99		0 0 0	0
0.0	0.0	273	75	89.0	
0.0		43.0		0.0	0
1.0	0 (19,[0,1,3,10,11,...	[26.9475640043703...		0.0	
33110	1	0 1.24		0 0 0	0
0.0	0.0	509	282	189.0	
0.0		178.0		0.0	0
2.0	0 (19,[0,1,3,10,11,...	[4.53601993227802...		0.0	

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only showing top 20 rows

```
[ ]:
```

```
[48]: from pyspark.ml.evaluation import BinaryClassificationEvaluator

bcv = BinaryClassificationEvaluator(rawPredictionCol='rawPrediction',
    metricName='areaUnderROC')
auc = bcv.evaluate(prediction.withColumnRenamed("sold_well", "label"))
print(auc)
```

```
0.9999926374705883
```

3 Revenue vs Positive Review Ratio - Linear Regression

Problem Statement: Is there a correlation between revenue and the ratio between the number of positive and negative reviews?

Algorithm Choice - Why Linear Regression? The problem asks specifically about *correlation* between a continuous predictor (positive review ratio) and a continuous outcome (estimated revenue). Linear Regression because R^2 directly quantifies how much of the variance in revenue is explained by the ratio, and the slope coefficient tells us the direction and magnitude of the relationship. K-Means is already used to address the related problem of revenue vs. total review count. Linear Regression gives a third different model to run on the data that better fits this analysis(I think).

3.1 Data Preparation

We compute `positive_ratio` (Positive / total reviews) as our feature and `log_revenue` (log-transformed estimated revenue) as our label. The log transform is necessary because revenue is extremely right-skewed — a handful of blockbuster titles would otherwise dominate the squared-error loss and make the model nearly meaningless for the majority of games. Games with zero revenue (free-to-play, no price signal) are filtered out, as are games with fewer than 10 total reviews — a game with 1 positive and 0 negative reviews has a ratio of 1.0 identical to a game with 10,000 positive reviews, which is just added noise to the model rather than a signal. Requiring at least 10 reviews ensures the ratio is statistically meaningful.

```
[49]: from pyspark.ml.feature import VectorAssembler

lr_data = data \
    .filter((functions.col('Positive') + functions.col('Negative')) >= 10) \
    .filter(functions.col('estimated_revenue') > 0) \
    .withColumn('positive_ratio',
        functions.col('Positive') / (functions.col('Positive') + functions.
    col('Negative'))) \
    .withColumn('log_revenue', functions.log1p(functions.
    col('estimated_revenue')))
```

```

assembler = VectorAssembler(
    inputCols=['positive_ratio'],
    outputCol='features',
    handleInvalid='keep'
)

lr_data = assembler.transform(lr_data)
lr_data.show()

```

```

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+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| AppId|          Name|Release date|Estimated owners|Peak CCU|Required
age|Price|DiscountDLC count| Supported languages|Full audio languages|Windows|
Mac|Linux|Metacritic score|User score|Positive|Negative|Average playtime
forever|Average playtime two weeks|Median playtime forever|Median playtime two
weeks|          Developers|          Publishers|          Categories|
Genres|          Tags|estimated_owners_mid|
estimated_revenue|num_languages|    positive_ratio|    log_revenue|
features|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| 496350|Supipara - Chapte...| 2016-07-29|    0 - 20000|    0|
0| 5.24|          65|    ['English']|    []|
NULL|NULL| NULL|    0.0|    0.0|    252|    3|
8.0|          0.0|          8.0|
0.0|          minori|    MangaGamer|['Single-player',...|
['Adventure']|['Adventure', 'Vi...|    NULL|    52400.0|
1.0|0.9882352941176471|10.866680954096502|[0.9882352941176471]|
|1034400|Mystery Solitaire...| 2019-05-06|    0 - 20000|    0|
0| 4.99|          0|['English', 'Fren...|    []|
NULL|NULL| NULL|    0.0|    0.0|    21|    3|
0.0|          0.0|          0.0|
0.0|    Somer Games|    8floor|['Single-player',...|
['Casual']|['Casual', 'Card ...|    NULL|    49900.0|
4.0|          0.875|10.817796321618971|    [0.875]|
|1934300| Armored Brigade II| 2025-04-08|    0 - 20000|    8|
0|35.99|          10|    ['English']|    []|
NULL|NULL| NULL|    0.0|    0.0|    117|    13|

```

675.0		0.0		1253.0	
0.0	Veitikka Studios		Matrix Games	['Single- player',...	['Simulation', 'S...
359900.0		1.0		0.9	12.79358427261286
[0.9]					
1157670	Hepta Beats	2021-05-07		0 - 20000	0
0 0.99		0	['English', 'Fren...	['English', 'Fren...	
NULL NULL	NULL	0.0	0.0	28	4
0.0		0.0		0.0	
0.0	Matheus Bigatão		Matheus Bigatão	['Single-player',...	['Casual', 'Indie']
	['Precision Platf...			NULL	9900.0
29.0		0.875	9.200391041122515		[0.875]
1540330	MUMBA IV: Egypt J...	2021-12-13		0 - 20000	1
0 0.59		0	['English', 'Fren...		[]
NULL NULL	NULL	0.0	0.0	19	6
0.0		0.0		0.0	
0.0	Airem		Airem	['Single-player',...	['Action', 'Casua...
	['Casual', 'Actio...			NULL	5900.0
30.0		0.76	8.682877107057168		[0.76]
1108640	OMNIMUS	2019-09-25		0 - 20000	0
0 0.99		0	['English', 'Russ...		[]
NULL NULL	NULL	0.0	0.0	47	6
0.0		0.0		0.0	
0.0	Angry games		Total	['Single-player',...	['Adventure', 'In...
	['Adventure', 'In...			NULL	9900.0
2.0 0.8867924528301887		9.200391041122515		[0.8867924528301887]	
1025440	Fantasy General II	2019-09-05		100000 - 200000	13
0 13.99		0	['English', 'Germ...		[]
NULL NULL	NULL	80.0	0.0	1025	210
489.0		0.0		52.0	
0.0	Owned by Gravity		Slitherine Ltd.	['Single-player',...	
	['Strategy']		['Strategy', 'Tur...		NULL
6.0 0.8299595141700404		14.556733838286524		[0.8299595141700404]	
2113450	Forests, Fields a...	2023-04-05		0 - 20000	0
0 1.49		70	['English', 'Russ...		[]
NULL NULL	NULL	0.0	0.0	64	2
0.0		0.0		0.0	
0.0	9 Eyes Game Studio		9 Eyes Game Studi...	['Single-player',...	['Casual', 'Indie']
	['Casual', 'Strat...			NULL	14900.0
2.0 0.9696969696969697		9.60918360377546		[0.9696969696969697]	
2922740	DogDogDog	2024-04-19		0 - 20000	0
0 1.35		66	['English', 'Simp...		[]
NULL NULL	NULL	0.0	0.0	34	10
0.0		0.0		0.0	
0.0	LuuuLuuuL		LuuuLuuuL	['Single-player',...	['Adventure', 'Ca...
	['Dog', 'Cute', '...			NULL	13500.0
4.0 0.7727272727272727		9.510519035757246		[0.7727272727272727]	
328460	Redline	2014-10-23		0 - 20000	0

0	1.04	0	['English']	74	35	203.0	0.0	0.0	203.0	0.0	Accolade, Inc. Atari, Nightdive S...	['Single-player', ...]	['Action', 'Racing']	['Action', 'Racin...]	NULL	10400.0	1.0	0.6788990825688074	9.249657234353133	[0.6788990825688074]		1607040	Evertried	2021-10-21	0 - 20000	0	0	5.99	0	['English', 'Fren...]	7	15	0.0	0.0	55	14	480.0	0.0	480.0	0.0	Lunic Games / Dan... DANGEN Entertainment	['Single-player', ...]	['Indie', 'Strate...']	['Strategy', 'Rog...]	NULL	59900.0	10.0	0.7971014492753623	11.000448478455008	[0.7971014492753623]		688420	Bad North: Jotunn...	2018-11-16	500000 - 1000000	64	0	2.99	80	['English', 'Fren...]	11367	831	342.0	86.0	173.0	139.0	Plausible Concept...	Raw Fury	['Single-player', ...]	['Action', 'Indie...']	['Wargame', 'Viki...]	NULL	2242500.0	12.0	0.9318740777176586	14.62310231884587	[0.9318740777176586]		496910	DrumKit VR - Play...	2016-07-08	0 - 20000	0	0	4.99	0	['English']	7	15	0.0	0.0	0.0	0.0	0o0	0o0	['Tracked Control...']	['Audio Production']	['Audio Productio...]	NULL	49900.0	1.0	0.3181818181818182	10.817796321618971	[0.3181818181818182]		2334500	Wizard Shrimp	2024-12-13	0 - 20000	1	0	2.99	0	['English']	95	2	0.0	0.0	0.0	0.0	0.0	Wizard Shrimp Games	Wizard Shrimp Games	['Single-player', ...]	['Action', 'Casua...']	['Action Roguelik...]	NULL	29900.0000000000004	1.0	0.979381443298969	10.305647203635566	[0.979381443298969]		1479370	Pixel Game Maker ...	2020-12-20	0 - 20000	0	0	9.99	0	['English', 'Simp...]	12	0	0.0	0.0	0.0	0.0	0.0	SERIALGAMES Inc.	Gotcha Gotcha Games	['Single-player', ...]	['Action', 'Indie']	['Action', 'Arcad...]	NULL	99900.0	4.0	1.0	11.511934974596555	[1.0]		2808780	OPERATION TEMPLE	2024-02-14	0 - 20000	0	0	5.99	80	['English', 'Russ...']	['English', 'Russ...']	NULL	10	2	0.0	0.0	0.0	0.0	Lieb	Lieb	['Single-player', ...]	['Action', 'Adven...']	['Adventure', 'Ac...]	NULL	59900.0	2.0	0.8333333333333334	11.000448478455008	[0.8333333333333334]
---	------	---	-------------	----	----	-------	-----	-----	-------	-----	---------------------------------------	------------------------	----------------------	-----------------------	------	---------	-----	--------------------	-------------------	----------------------	--	---------	-----------	------------	-----------	---	---	------	---	-----------------------	---	----	-----	-----	----	----	-------	-----	-------	-----	--	------------------------	------------------------	-----------------------	------	---------	------	--------------------	--------------------	----------------------	--	--------	----------------------	------------	------------------	----	---	------	----	-----------------------	-------	-----	-------	------	-------	-------	----------------------	----------	------------------------	------------------------	-----------------------	------	-----------	------	--------------------	-------------------	----------------------	--	--------	----------------------	------------	-----------	---	---	------	---	-------------	---	----	-----	-----	-----	-----	-----	-----	------------------------	----------------------	-----------------------	------	---------	-----	--------------------	--------------------	----------------------	--	---------	---------------	------------	-----------	---	---	------	---	-------------	----	---	-----	-----	-----	-----	-----	---------------------	---------------------	------------------------	------------------------	-----------------------	------	---------------------	-----	-------------------	--------------------	---------------------	--	---------	----------------------	------------	-----------	---	---	------	---	-----------------------	----	---	-----	-----	-----	-----	-----	------------------	---------------------	------------------------	---------------------	-----------------------	------	---------	-----	-----	--------------------	-------	--	---------	------------------	------------	-----------	---	---	------	----	------------------------	------------------------	------	----	---	-----	-----	-----	-----	------	------	------------------------	------------------------	-----------------------	------	---------	-----	--------------------	--------------------	----------------------

26/04/26 21:33:59 WARN Instrumentation: [e5118ce6] regParam is zero, which might cause numerical instability and overfitting.

```
[52]: prediction = model.transform(test)
      prediction.show()
```

```
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+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|AppId|          Name|Release date|Estimated owners|Peak CCU|Required
age|Price|DiscountDLC count| Supported languages|Full audio languages|Windows|
Mac|Linux|Metacritic score|User score|Positive|Negative|Average playtime
forever|Average playtime two weeks|Median playtime forever|Median playtime two
weeks|          Developers|          Publishers|          Categories|
Genres|          Tags|estimated_owners_mid|
estimated_revenue|num_languages|    positive_ratio|    log_revenue|
features|    prediction|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+
| 2210|          Quake 4|  2011-08-04| 200000 - 500000|    18|    17|
4.94|          0|    ['English']|    []|    NULL|NULL|
NULL|          81.0|    0.0|    3001|    423|    160.0|
46.0|          66.0|          46.0|RavenSoft,id Soft...|
Bethesda Softworks|['Single-player',...|    ['Action']|['FPS',
'Action',...|    NULL|1729000.0000000002|
1.0|0.8764602803738317|
14.36305434303426|[0.8764602803738317]|11.427553464107739|
| 3510|Amazing Adventure...|  2007-12-13|  20000 - 50000|    4|    0|
0.99|          0|    ['English']|    []|    NULL|NULL|
NULL|          0.0|    0.0|    70|    17|    0.0|
0.0|          0.0|          0.0|    0.0|    SpinTop Games|
PopCap Games, Inc.|['Single-player',...|    ['Casual']|['Casual',
'Hide...|    NULL|    34650.0|
1.0|0.8045977011494253|10.453081864230466|[0.8045977011494253]|
11.34430240716659|
| 3610|Mystery P.I. - Lo...|  2009-07-27|  20000 - 50000|    5|    0|
0.99|          0|    ['English']|    []|    NULL|NULL|
NULL|          0.0|    0.0|    43|    11|    13.0|
0.0|          13.0|          0.0|    0.0|    SpinTop Games|
```

```

None|['Single-player',...|          ['Casual']]|['Casual', 'Hidde...|
NULL|          34650.0|
1.0|0.7962962962962963|10.453081864230466|[0.7962962962962963]|
11.33468543077111|
| 6030|STAR WARS Jedi K...| 2009-09-16|500000 - 1000000|          40|          0|
4.99|          50|['English', 'Fren...|['English', 'Fren...| NULL|NULL|
NULL|          89.0|          0.0|          4281|          413|          392.0|
0.0|          117.0|          0.0|          Raven
Software|LucasArts, Lucasfi...|['Single-player',...|
['Action']]|['Action', 'Class...|          NULL|          3742500.0|
5.0| 0.912015338730294|15.135264662476954|
[0.912015338730294]|11.468743137783887|
| 7860|          NecroVision| 2009-05-20| 200000 - 500000|          0|          17|
0.74|          85|['English', 'Fren...|['English', 'Fren...| NULL|NULL|
NULL|          63.0|          0.0|          1181|          554|          268.0|
0.0|          119.0|          0.0|          The Farm 51|
Fulqrum Publishing|['Single-player',...|          ['Action']]|['Action',
'FPS',...|          NULL|          259000.0|
9.0|0.6806916426512968|12.464587201678082|[0.6806916426512968]|
11.20076024039897|
| 8800|Civilization IV: ...| 2007-07-24|500000 - 1000000|          1395|          0|
5.99|          80|['English', 'Fren...|          []| NULL|NULL|
NULL|          86.0|          0.0|          4423|          169|          2378.0|
1661.0|          258.0|          1661.0|          Firaxis Games|
2K|['Single-player',...|          ['Strategy']]|['Strategy', 'Tur...|
NULL|          4492500.0|
5.0|0.9631968641114983|15.317920120233037|[0.9631968641114983]|
11.52803569490934|
| 8880|          Freedom Force| 2009-05-29| 200000 - 500000|          1|          0|
1.24|          0|          ['English']]|          []| NULL|NULL|
NULL|          90.0|          0.0|          292|          230|          629.0|
0.0|          78.0|          0.0|          Irrational Games|
2K|['Single-player',...| ['RPG', 'Strategy']]|['RPG', 'Strategy...|
NULL|          434000.0|          1.0|0.5593869731800766|12.980802117227352|[0.5
593869731800766]|11.060231719792435|
|12520|18 Wheels of Stee...| 2008-01-29| 50000 - 100000|          3|          0|
4.49|          0|          ['English']]|          []| NULL|NULL|
NULL|          0.0|          0.0|          360|          134|          332.0|
0.0|          295.0|          0.0|          SCS Software|
Cosmi Valusoft|['Single-player',...|          ['Racing']]|['Automobile Sim'...|
NULL|          336750.0|          1.0| 0.728744939271255|12.727099063830192|
[0.728744939271255]|11.256428820295408|
|12640|          Drakensang| 2009-03-09| 100000 - 200000|          6|          0|
1.99|          0|['English', 'Germ...|          []| NULL|NULL|
NULL|          75.0|          0.0|          592|          143|          1018.0|
9.0|          254.0|          9.0|          Radon Labs|
DTP Entertainment|['Single-player',...|          ['RPG']]|['RPG',
'Fantasy'...|          NULL|          298500.0|

```

```

3.0|0.8054421768707483|12.606528561892935|[0.8054421768707483]|
11.34528071182974|
|13570|Tom Clancy's Spli...| 2009-08-27| 200000 - 500000|      26|      17|
2.49|      0|['English', 'Fren...|      []| NULL|NULL|
NULL|      92.0|      0.0|      4785|      348|      523.0|
0.0|      266.0|      0.0|      Ubisoft Montreal|
Ubisoft|['Single-player',...|      ['Action']|['Stealth', 'Acti...|
NULL| 871500.0000000001|      5.0|0.9322033898305084|13.677972291388485|[0.9
322033898305084]|11.492130505866328|
|13600|Prince of Persia®...| 2008-11-21|500000 - 1000000|      26|      0|
1.99|      0|['English', 'Fren...|      []| NULL|NULL|
NULL|      89.0|      0.0|      3844|      637|      252.0|
146.0|      88.0|      146.0|      Kudosoft|
Ubisoft|['Single-player',...|['Action', 'Adven...|['Action', 'Adven...|
NULL|      1492500.0|      5.0|0.8578442311983933|
14.21596379426542|[0.8578442311983933]|11.405987222168605|
|16200|Virtual Families| 2009-05-06|      0 - 20000|      2|      0|
9.99|      0|      ['English']|      []| NULL|NULL|
NULL|      0.0|      0.0|      58|      9|      807.0|
0.0|      1026.0|      0.0|      Last Day of Work|
None|['Single-player',...|['Casual', 'Simul...|['Simulation', 'C...|
NULL|      99900.0|
1.0|0.8656716417910447|11.511934974596555|[0.8656716417910447]|
11.41505508769728|
|18000|Precipice of Dark...| 2008-05-21| 100000 - 200000|      0|      17|
7.99|      0|      ['English']|      []| NULL|NULL|
NULL|      0.0|      0.0|      451|      58|      376.0|
0.0|      139.0|      0.0|      Hothead Games|
Hothead Games|['Single-player',...|['Adventure', 'RPG']|['RPG', 'Adventur...|
NULL|      1198500.0|      1.0|0.8860510805500982|13.996582167232532|[0.8
860510805500982]|11.438664173745535|
|18820|Zero Gear| 2010-01-12| 50000 - 100000|      0|      0|
1.24|      0|      ['English']|      []| NULL|NULL|
NULL|      0.0|      0.0|      687|      189|      396.0|
0.0|      86.0|      0.0|      Brian Cronin|
Brian Cronin|['Single-player',...|['Action', 'Indie...|['Racing', 'Indie...|
NULL|      93000.0|      1.0|0.7842465753424658|11.440365524765756|[0.7
842465753424658]|11.320726120987366|
|31800|Nancy Drew®: Dang...| 2009-10-10| 20000 - 50000|      6|      0|
6.49|      0|      ['English']|      []| NULL|NULL|
NULL|      70.0|      0.0|      129|      27|      391.0|
0.0|      397.0|      0.0|      HeR Interactive|
HeR Interactive|['Single-player',...|      ['Adventure']|['Adventure', 'Po...|
NULL|      227150.0|      1.0|0.8269230769230769|12.333370273555143|[0.8
269230769230769]|11.370165814099993|
|33110|Alien Shooter: Re...| 2009-05-27| 200000 - 500000|      1|      0|
1.24|      0|['English', 'Russ...|['English', 'Russ...| NULL|NULL|
NULL|      0.0|      0.0|      509|      282|      189.0|

```

```

0.0|          178.0|          0.0|      Sigma Team Inc.|
Sigma Team Inc.|['Single-player',...|['Action', 'Indie...|['Action', 'RPG',...|
NULL|          434000.0|          2.0|0.6434892541087232|12.980802117227352|[0.6
434892541087232]|11.157662174613813|
|33390|          Zeit²|  2011-01-12|          0 - 20000|          0|          0|
1.99|          80|['English', 'Fren...|          []|      NULL|NULL|
NULL|          72.0|          0.0|          34|          7|          177.0|
0.0|          349.0|          0.0|      Brightside Games|
Ubisoft|['Single-player',...| ['Action', 'Indie']|["['Action', 'Indi...|
NULL|          10000.0|          19900.0|0.8292682926829268|
9.210440366976517|[0.8292682926829268]| 11.37288268973777|
|35320|          Insane 2|  2012-01-24|  50000 - 100000|          3|          0|
4.49|          0|['English', 'Russ...|['English', 'Russ...|      NULL|NULL|
NULL|          0.0|          0.0|          515|          103|          296.0|
0.0|          309.0|          0.0|      Targem Games|Game
Factory Inte...|['Single-player',...|          ['Racing']|['Racing', 'Offro...|
NULL|          336750.0|          2.0|0.8333333333333334|12.727099063830192|[0.8
3333333333333334]|11.377591940843248|
|37280|Wedding Dash® 2: ...|  2009-08-12|          0 - 20000|          7|          0|
9.99|          0|          ['English']|          []|      NULL|NULL|
NULL|          0.0|          0.0|          64|          19|          0.0|
0.0|          0.0|          0.0|          0.0|          Sarbakan|
None|['Single-player',...|          ['Simulation']|['Simulation', 'T...|
NULL|          99900.0|          1.0|0.7710843373493976|11.511934974596555|[0.7
710843373493976]|11.305477987167787|
|37390|          Gemini Lost |  2010-05-06|          0 - 20000|          0|          0|
6.99|          0|          ['English']|          []|      NULL|NULL|
NULL|          0.0|          0.0|          51|          1|          1398.0|
0.0|          1398.0|          0.0|          0.0|          PlayFirst|
None|['Single-player',...|          ['Casual']|['Casual', 'Famil...|
NULL|          69900.0|          1.0|0.9807692307692307|11.154835234271214|[0.9
807692307692307]|11.548392855938104|
+-----+-----+-----+-----+-----+-----+
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```

only showing top 20 rows

```

[53]: from pyspark.ml.evaluation import RegressionEvaluator

evaluator_rmse = RegressionEvaluator(labelCol='log_revenue',
    ↪predictionCol='prediction', metricName='rmse')
evaluator_r2    = RegressionEvaluator(labelCol='log_revenue',
    ↪predictionCol='prediction', metricName='r2')

```

```

baseline_rmse = evaluator_rmse.evaluate(prediction)
baseline_r2    = evaluator_r2.evaluate(prediction)

print("--- Baseline ---")
print("RMSE: " + str(round(baseline_rmse, 4)))
print("R2:   " + str(round(baseline_r2, 4)))
print("Coefficient: " + str(round(model.coefficients[0], 4)))
print("Intercept:   " + str(round(model.intercept, 4)))

```

[Stage 2433:=====> (12 + 2) / 14]

```

--- Baseline ---
RMSE: 1.7584
R2:   0.0101
Coefficient: 1.1585
Intercept:  10.4122

```

3.3 Hyperparameter Tuning — Grid Search

We use `ParamGridBuilder` with `TrainValidationSplit` to perform a grid search over two parameters:

- **regParam** (): regularization strength. 0.0 = no regularization (plain OLS), higher values shrink the coefficient to reduce overfitting.
- **elasticNetParam** (): controls the L1/L2 mix. 0.0 = Ridge, 1.0 = Lasso.

This gives 9 combinations (3×3). `TrainValidationSplit` evaluates each on an 80/20 internal split of the training data using RMSE and selects the best.

[54]: `from pyspark.ml.tuning import ParamGridBuilder, TrainValidationSplit`

```

paramGrid = ParamGridBuilder() \
    .addGrid(lr.regParam,      [0.0, 0.01, 0.1]) \
    .addGrid(lr.elasticNetParam, [0.0, 0.5, 1.0]) \
    .build()

tvsv = TrainValidationSplit(
    estimator=lr,
    estimatorParamMaps=paramGrid,
    evaluator=evaluator_rmse,
    trainRatio=0.8,
    seed=random_seed
)

best_model      = tvsv.fit(train).bestModel
best_prediction = best_model.transform(test)

best_rmse = evaluator_rmse.evaluate(best_prediction)

```

```

best_r2    = evaluator_r2.evaluate(best_prediction)

print("--- Best Model (after grid search) ---")
print("regParam:      " + str(best_model.getRegParam()))
print("elasticNetParam: " + str(best_model.getElasticNetParam()))
print("RMSE: " + str(round(best_rmse, 4)))
print("R2:    " + str(round(best_r2, 4)))
print("Coefficient: " + str(round(best_model.coefficients[0], 4)))
print("Improvement in RMSE over baseline: " + str(round(baseline_rmse -
↳best_rmse, 4)))

```

26/04/26 21:34:07 WARN Instrumentation: [3baf65d4] regParam is zero, which might cause numerical instability and overfitting.

26/04/26 21:34:10 WARN Instrumentation: [38cddf08] regParam is zero, which might cause numerical instability and overfitting.

26/04/26 21:34:11 WARN Instrumentation: [2892159a] regParam is zero, which might cause numerical instability and overfitting.

[Stage 2526:===> (1 + 13) / 14]

```

--- Best Model (after grid search) ---
regParam:      0.01
elasticNetParam: 0.5
RMSE: 1.7583
R2:    0.0102
Coefficient: 1.1279
Improvement in RMSE over baseline: 0.0001

```

[]:

3.4 Analysis and Visualization

The scatter plot below shows each game in the test set as a point (positive_ratio vs log_revenue), with the fitted regression line overlaid. Steam games tend to cluster heavily toward high positive ratios (0.7–1.0), so expect a dense cloud on the right side. The Pearson correlation is reported as a model-free confirmation — if it agrees in sign with the regression coefficient, the linear relationship is genuine rather than a modeling artifact.

Interpreting the results: The R^2 value tells us directly how strong the correlation is. A positive coefficient means games with better review ratios tend to earn more revenue. However, a low R^2 (which is expected here — review ratio is just one of many factors that determine revenue) does not mean the correlation is absent; it means review quality alone is not a complete predictor of commercial success, which is itself an informative finding.

An Interesting Result: When run with 100 the grid search picked regParam=0.1, elasticNetParam=0.5 instead of no regularization. That means with the smaller, noisier dataset (≥ 10) we actually needed to shrink the coefficient to avoid overfitting, which I guess shows the tuning doing exactly what it's supposed to.

[]:

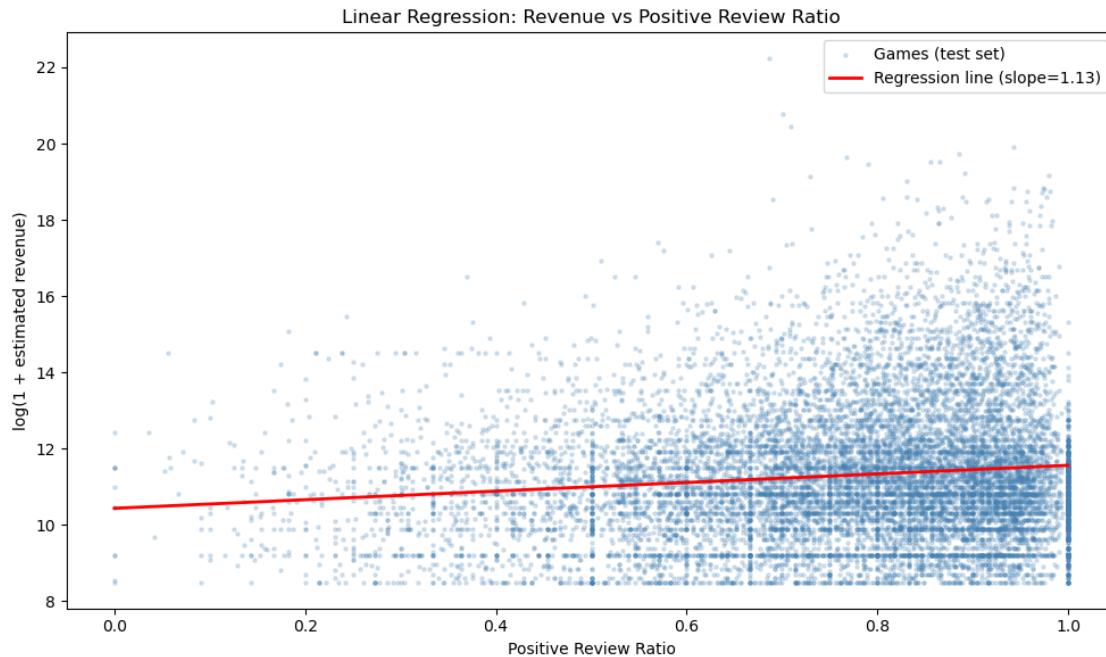
```
[55]: import matplotlib.pyplot as plt
import numpy as np

plot_df = best_prediction.select('positive_ratio', 'log_revenue', 'prediction').
    ↪toPandas()

coeff      = best_model.coefficients[0]
intercept  = best_model.intercept
x_line     = np.linspace(plot_df['positive_ratio'].min(),
    ↪plot_df['positive_ratio'].max(), 200)
y_line     = coeff * x_line + intercept

plt.figure(figsize=(10, 6))
plt.scatter(plot_df['positive_ratio'], plot_df['log_revenue'],
            alpha=0.2, s=5, color='steelblue', label='Games (test set)')
plt.plot(x_line, y_line, color='red', linewidth=2,
        label='Regression line (slope=' + str(round(coeff, 2)) + ')')
plt.xlabel('Positive Review Ratio')
plt.ylabel('log(1 + estimated revenue)')
plt.title('Linear Regression: Revenue vs Positive Review Ratio')
plt.legend()
plt.tight_layout()
plt.savefig('revenue_ratio_regression.png', dpi=120)
plt.show()

pearson = lr_data.stat.corr('positive_ratio', 'log_revenue')
print("Pearson correlation: " + str(round(pearson, 4)))
print("R2: " + str(round(best_r2, 4)))
print("The positive ratio explains " + str(round(best_r2 * 100, 2)) + "% of
    ↪variance in log revenue")
```

Pearson correlation: 0.1132

R2: 0.0102

The positive ratio explains 1.02% of variance in log revenue

4 Revenue vs Number of Reviews - K-Means Clustering

4.1 Data Preparation

```
[56]: from pyspark.ml.feature import VectorAssembler, StandardScaler

km_data = data \
    .filter((functions.col('Positive') + functions.col('Negative')) > 0) \
    .filter(functions.col('estimated_revenue') > 0) \
    .withColumn('total_reviews', functions.col('Positive') + functions.
    ↪ col('Negative')) \
    .withColumn('log_revenue', functions.log1p(functions.
    ↪ col('estimated_revenue'))) \
    .withColumn('log_reviews', functions.log1p(functions.
    ↪ col('total_reviews')))

assembler = VectorAssembler(
    inputCols=['log_revenue', 'log_reviews'],
    outputCol='raw_features',
    handleInvalid='keep'
)
```

```

km_data = assembler.transform(km_data)

scaler = StandardScaler(
    inputCol='raw_features',
    outputCol='features',
    withMean=True,
    withStd=True
)
km_data = scaler.fit(km_data).transform(km_data)
km_data.show()

```

```

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+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+
| AppId|          Name|Release date|Estimated owners|Peak CCU|Required
age|Price|DiscountDLC count| Supported languages|Full audio languages|Windows|
Mac|Linux|Metacritic score|User score|Positive|Negative|Average playtime
forever|Average playtime two weeks|Median playtime forever|Median playtime two
weeks|          Developers|          Publishers|          Categories|
Genres|          Tags|estimated_owners_mid|
estimated_revenue|num_languages|total_reviews|          log_revenue|
log_reviews|          raw_features|          features|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+
| 496350|Supipara - Chapte...| 2016-07-29|          0 - 20000|          0|
0| 5.24|          65|          ['English']|          []|
NULL|NULL| NULL|          0.0|          0.0|          252|          3|
8.0|          0.0|          8.0|
0.0|          minori|          MangaGamer|['Single-player',...|
['Adventure']|['Adventure', 'Vi...|          NULL|          52400.0|
1.0|          255|10.866680954096502|
5.545177444479562|[10.8666809540965...|[-0.0965837046732...|
|1034400|Mystery Solitaire...| 2019-05-06|          0 - 20000|          0|
0| 4.99|          0|['English', 'Fren...|          []|
NULL|NULL| NULL|          0.0|          0.0|          21|          3|
0.0|          0.0|          0.0|
0.0|          Somer Games|          8floor|['Single-player',...|

```

```

['Casual']|['Casual', 'Card ...|          NULL|          49900.0|
4.0|          24|10.817796321618971|3.2188758248682006|[10.8177963216189...|[-
0.1264412730679...|
|1934300| Armored Brigade II| 2025-04-08|          0 - 20000|          8|
0|35.99|          10|          ['English']|          []|
NULL|NULL| NULL|          0.0|          0.0|          117|          13|
675.0|          0.0|          1253.0|
0.0| Veitikka Studios| Matrix Games|['Single-
player',...|['Simulation', 'S...|['Simulation', 'S...|          NULL|
359900.0|          1.0|          130| 12.79358427261286|
4.875197323201151|[12.7935842726128...|[1.08032291621929...|
|1157670| Hepta Beats| 2021-05-07|          0 - 20000|          0|
0| 0.99|          0|['English', 'Fren...|['English', 'Fren...|
NULL|NULL| NULL|          0.0|          0.0|          28|          4|
0.0|          0.0|          0.0|
0.0| Matheus Bigatão| Matheus Bigatão|['Single-player',...| ['Casual',
'Indie']|['Precision Platf...|          NULL|          9900.0|
29.0|          32|
9.200391041122515|3.4965075614664802|[9.20039104112251...|[-1.1143138689532...|
|1540330|MUMBA IV: Egypt J...| 2021-12-13|          0 - 20000|          1|
0| 0.59|          0|['English', 'Fren...|          []|
NULL|NULL| NULL|          0.0|          0.0|          19|          6|
0.0|          0.0|          0.0|
0.0| Airem| Airem|['Single-player',...| ['Action',
'Casua...|['Casual', 'Actio...|          NULL|          5900.0|
30.0|          25| 8.682877107057168|
3.258096538021482|[8.68287710705716...|[-1.4303990453878...|
|1108640| OMNIMUS| 2019-09-25|          0 - 20000|          0|
0| 0.99|          0|['English', 'Russ...|          []|
NULL|NULL| NULL|          0.0|          0.0|          47|          6|
0.0|          0.0|          0.0|
0.0| Angry games| Total|['Single-player',...| ['Adventure',
'In...|['Adventure', 'In...|          NULL|          9900.0|
2.0|          53|
9.200391041122515|3.9889840465642745|[9.20039104112251...|[-1.1143138689532...|
|1025440| Fantasy General II| 2019-09-05| 100000 - 200000|          13|
0|13.99|          0|['English', 'Germ...|          []|
NULL|NULL| NULL|          80.0|          0.0|          1025|          210|
489.0|          0.0|          52.0|
0.0| Owned by Gravity| Slitherine Ltd.|['Single-player',...|
['Strategy']|['Strategy', 'Tur...|          NULL|          2098500.0|
6.0|          1235|14.556733838286524|
7.119635638017636|[14.5567338382865...|[2.15721264780313...|
|2113450|Forests, Fields a...| 2023-04-05|          0 - 20000|          0|
0| 1.49|          70|['English', 'Russ...|          []|
NULL|NULL| NULL|          0.0|          0.0|          64|          2|
0.0|          0.0|          0.0|
0.0| 9 Eyes Game Studio|9 Eyes Game Studi...|['Single-player',...| ['Casual',

```

'Indie'] ['Casual', 'Strat...	NULL	14900.0
2.0	66 9.60918360377546	
4.204692619390966	[9.60918360377546...	[-0.8646331148984...
2922740	DogDogDog 2024-04-19	0 - 20000 0
0 1.35	66 ['English', 'Simp...	[]
NULL NULL NULL	0.0 0.0	34 10
0.0	0.0	0.0
0.0	LuuuLuuuL	LuuuLuuuL ['Single-player',...
'Ca... ['Dog', 'Cute', '...	NULL	13500.0
4.0	44	
9.510519035757246	3.8066624897703196	[9.51051903575724...
[-0.9248950814602...		
2517090	Challenge from th... 2023-08-10	0 - 20000 0
0 1.4	0 ['English', 'Japa...	['Japanese', 'Kor...
NULL NULL NULL	0.0 0.0	2 0
0.0	0.0	0.0
0.0	ZENO	ZENO ['Single-player',...
['Action'] ['Action', 'Shoot...	NULL	14000.0
13.0	2	
9.546884034617925	1.0986122886681096	[9.54688403461792...
[-0.9026842068840...		
328460	Redline 2014-10-23	0 - 20000 0
0 1.04	0	['English']
NULL NULL NULL	0.0 0.0	74 35
203.0	0.0	203.0
0.0	Accolade, Inc. Atari,Nightdive S...	['Single-player',...
'Racing'] ['Action', 'Racin...	NULL	10400.0
1.0	109 9.249657234353133	
4.700480365792417	[9.24965723435313...	[-1.0842232523446...
1607040	Evertried 2021-10-21	0 - 20000 0
0 5.99	0 ['English', 'Fren...	[]
NULL NULL NULL	0.0 0.0	55 14
480.0	0.0	480.0
0.0	Lunic Games / Dan... DANGEN Entertainment	['Single-player',...
'Strate... ['Strategy', 'Rog...	NULL	59900.0
10.0	69 11.000448478455008	
4.248495242049359	[11.0004484784550...	[-0.0148816890271...
815170	Zombie World 2018-06-01	0 - 20000 0
0 7.99	0 ['English', 'Simp...	['English', 'Kore...
NULL NULL NULL	0.0 0.0	0 4
0.0	0.0	0.0
0.0	M.R.Studio	M.R.Studio ['Single-player',...
'Adven... ['Indie', 'Action...	NULL	79900.0
8.0	4 11.288543647320601	1.6094379124341003
[11.2885436473206...	[0.16107997361931...	
688420	Bad North: Jotunn... 2018-11-16	500000 - 1000000 64
0 2.99	80 ['English', 'Fren...	[]
NULL NULL NULL	0.0 0.0	11367 831
342.0	86.0	173.0
139.0	Plausible Concept...	Raw Fury ['Single-player',...
		['Action',

```

'Indie...|['Wargame', 'Viki...|          NULL|          2242500.0|
12.0|          12198| 14.62310231884587|
9.409109260148737|[14.6231023188458...|[2.19774893389402...|
|3281190|They Yearn For Th...| 2024-12-04|          0 - 20000|          0|
0| 5.99|          0|          ['English']|          ['English']|
NULL|NULL| NULL|          0.0|          0.0|          6|          0|
0.0|          0.0|          0.0|
0.0|          TheQuokkaBloke|          TheQuokkaBloke|['Single-player',...|['Action',
'Adven...|['Adventure', 'Sp...|          NULL|          59900.0|
1.0|          6|11.000448478455008|1.9459101490553132|[11.0004484784550...|[-
0.0148816890271...|
|2260360|          Zooblies| 2023-01-16|          0 - 20000|          0|
0| 1.99|          0|          ['English']|          []|
NULL|NULL| NULL|          0.0|          0.0|          5|          0|
0.0|          0.0|          0.0|
0.0|Jolly Crouton Med...|Jolly Crouton Med...|['Single-player',...|['Action',
'Casual']|['Puzzle-Platform...|          NULL|          19900.0|
1.0|          5| 9.898525260706313|
1.791759469228055|[9.89852526070631...|[-0.6879101270507...|
| 496910|DrumKit VR - Play...| 2016-07-08|          0 - 20000|          0|
0| 4.99|          0|          ['English']|          []|
NULL|NULL| NULL|          0.0|          0.0|          7|          15|
0.0|          0.0|          0.0|
0.0|          0o0|          0o0|['Tracked Control...|['Audio
Production']|['Audio Productio...|          NULL|          49900.0|
1.0|          22|10.817796321618971|3.1354942159291497|[10.8177963216189...|[-
0.1264412730679...|
|2334500|          Wizard Shrimp| 2024-12-13|          0 - 20000|          1|
0| 2.99|          0|          ['English']|          []|
NULL|NULL| NULL|          0.0|          0.0|          95|          2|
0.0|          0.0|          0.0|
0.0| Wizard Shrimp Games| Wizard Shrimp Games|['Single-player',...|['Action',
'Casua...|['Action Roguelik...|          NULL|29900.000000000004|
1.0|          97|10.305647203635566|
4.584967478670572|[10.3056472036355...|[-0.4392497477054...|
|2075820|          Controller Sync| 2023-11-18| 20000 - 50000|          0|
0| 0.99|          0|          ['English']|          ['English']|
NULL|NULL| NULL|          0.0|          0.0|          5|          0|
0.0|          0.0|          0.0|
0.0|          Matti Kuehl|          Matti Kuehl|['Single-player',...|['Action',
'Casua...|['Action', 'Arcad...|          NULL|          34650.0|
1.0|          5|10.453081864230466|
1.791759469228055|[10.4530818642304...|[-0.3492001709903...|
|1479370|Pixel Game Maker ...| 2020-12-20|          0 - 20000|          0|
0| 9.99|          0|['English', 'Simp...|          []|
NULL|NULL| NULL|          0.0|          0.0|          12|          0|
0.0|          0.0|          0.0|
0.0|          SERIALGAMES Inc.| Gotcha Gotcha Games|['Single-player',...| ['Action',

```


4.3 Analysis and Visualization

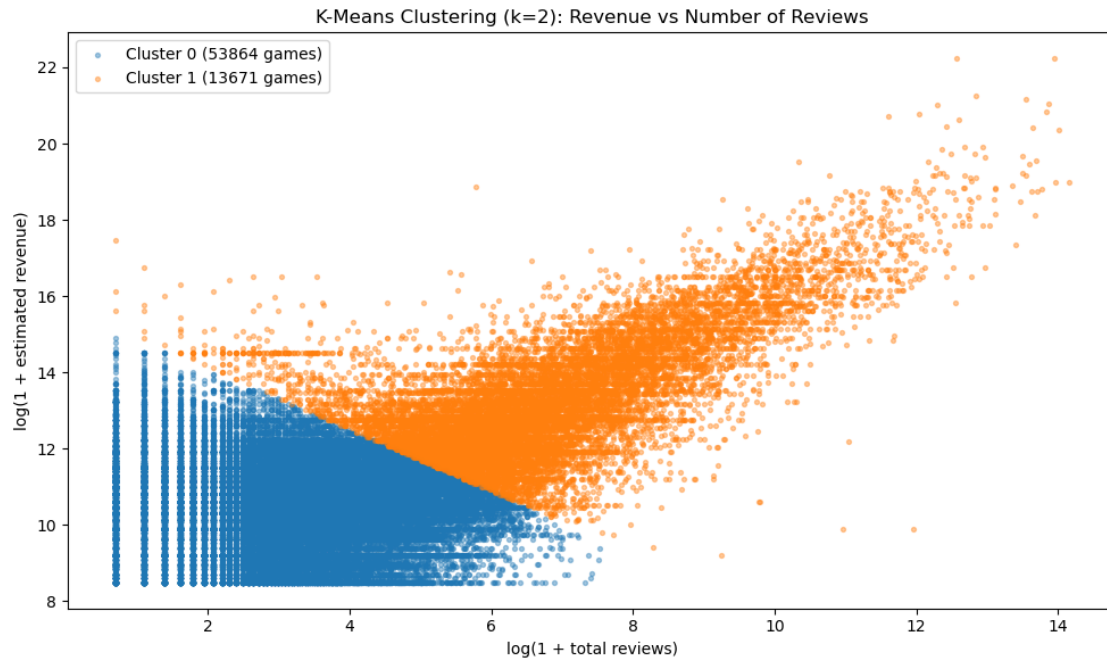
```
[58]: import matplotlib.pyplot as plt

plot_df = prediction.select('log_revenue', 'log_reviews', 'prediction').
    ↪toPandas()
k = model.getK()

plt.figure(figsize=(10, 6))
for c in range(k):
    sub = plot_df[plot_df['prediction'] == c]
    plt.scatter(sub['log_reviews'], sub['log_revenue'],
                alpha=0.4, s=8,
                label='Cluster ' + str(c) + ' (' + str(len(sub)) + ' games)')
plt.xlabel('log(1 + total reviews)')
plt.ylabel('log(1 + estimated revenue)')
plt.title('K-Means Clustering (k=' + str(k) + '): Revenue vs Number of Reviews')
plt.legend()
plt.tight_layout()
plt.savefig('revenue_reviews_clusters.png', dpi=120)
plt.show()

overall_corr = prediction.stat.corr('log_revenue', 'log_reviews')
print("Overall Pearson correlation (log scale): " + str(round(overall_corr, 4)))

for c in range(k):
    cluster_subset = prediction.filter(functions.col('prediction') == c)
    cluster_corr = cluster_subset.stat.corr('log_revenue', 'log_reviews')
    cluster_count = cluster_subset.count()
    print("Cluster " + str(c) + ": correlation=" + str(round(cluster_corr, 4)) +
          ", games=" + str(cluster_count))
```



Overall Pearson correlation (log scale): 0.6262

Cluster 0: correlation=-0.0199, games=53864

Cluster 1: correlation=0.6429, games=13671

```
[62]: spark.stop()
```

```
[ ]:
```